



Lesson 1

Introduction to Computers and Programming



What is in this lesson?

Hardware and Software

How Computers Store Data

How a Program Works

Getting Started: Python

What do you use computers for?

What do you use computers for?

- ▶ Browsing the web and communicating
- ▶ Studying and working
- ▶ Entertaining (playing games, watching videos, listening to music)
- ▶ Solving problems and automating tasks

Devices that are computers...

Devices that are computers...

- ▶ Laptops
- ▶ Smart phones
- ▶ iPods
- ▶ Car navigation system (GPS)

Computers are designed to do the jobs that their **programs** tell them to do.

What is a Program?

What is a Program?

A **Program** is a sequence of instructions that a computer follows in order to perform a task.

Hardware and Software

Hardware and Software

Hardware

The physical devices that a computer is made of are referred to as the computer's hardware.

Software

The programs that run on a computer are referred to as software.

Hardware

A computer is a system of devices that work together. A Computer System consists of:

- ▶ Central Processing Unit (CPU)
- ▶ Main memory
- ▶ Secondary storage
- ▶ Input devices
- ▶ Output devices

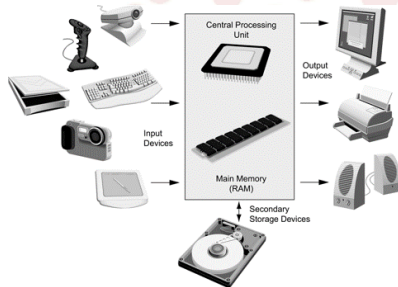


Figure: Typical components of a computer system

Central Processing Unit (CPU)

- ▶ The CPU is the part of a computer that runs the programs.
- ▶ Without a CPU, a computer cannot run software.

Run or **execute**: is the term used when the computer performs the tasks that the program tells it to do.

Computers Then and Now

ENIAC

- ▶ World's first programmable computer
- ▶ Built in 1945
- ▶ Designed to calculate artillery ballistic tables for the U.S. Army
- ▶ CPU was 8 feet tall, 100 feet long, and weighed 30 tons

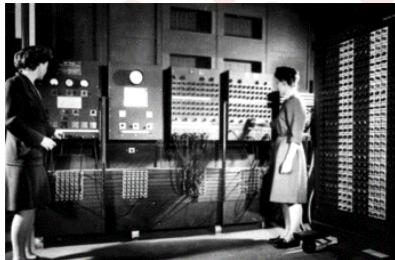


Figure: The ENIAC computer (courtesy of U.S. Army Historic Computer Images)

Microprocessor

- ▶ Much smaller
- ▶ Much more powerful



Figure: A lab technician holds a modern microprocessor

Main Memory

- ▶ Considered the computer's working area
- ▶ Computer stores the program that is running as well as the data
- ▶ Commonly known as the random-access memory (RAM)
- ▶ Data is quickly accessed
- ▶ RAM is a volatile type of memory
- ▶ Used for temporary storage
- ▶ RAM is erased when computer is turned off

Secondary Storage Devices

- ▶ Type of memory that can hold data for long periods of time
- ▶ Programs and important data are stored in secondary storage
- ▶ **Disk drive** is a common type of secondary storage
 - ▶ Data is stored by magnetically encoding it onto a circular disk
 - ▶ Most computers have an internal disk drive
 - ▶ Some have external disk drives; they are used to create backup copies
- ▶ **Floppy drives** record data onto a small floppy disk
 - ▶ Holds only a small amount of data
 - ▶ Slow to access data
 - ▶ Can be unreliable

- ▶ **USB drives** are small devices that plug into the computer's universal serial bus (USB) port
 - ▶ It does not contain a disk
 - ▶ The data is stored on flash memory
 - ▶ Also known as memory sticks and flash drives
 - ▶ Inexpensive, reliable, and small
- ▶ **Optical devices (CD or DVD)**
 - ▶ Data is encoded as a series of pits on the disc's surface
 - ▶ Uses laser to encode the data
 - ▶ Holds large amounts of data
 - ▶ Good medium for creating backups

Input Devices

- ▶ Any data the computer collects from humans and from other devices is called **input**.
- ▶ The hardware component that collects the data is called an **input device**.
- ▶ Common input devices include:

Input Devices

- ▶ Any data the computer collects from humans and from other devices is called **input**.
- ▶ The hardware component that collects the data is called an **input device**.
- ▶ Common input devices include:
 - ▶ Keyboard
 - ▶ Mouse
 - ▶ Scanner
 - ▶ Microphone
 - ▶ Digital camera

Output Devices

- ▶ Any data the computer produces for humans or for other devices is called **output**.
- ▶ The hardware component that formats and presents the data is called an **output device**.
- ▶ Common output devices include:

Ouput Devices

- ▶ Any data the computer produces for humans or for other devices is called **output**.
- ▶ The hardware component that formats and presents the data is called an **output device**.
- ▶ Common output devices include:
 - ▶ Monitor
 - ▶ Printer
 - ▶ Speaker

Software

- ▶ Everything a computer does is controlled by software.
- ▶ Two categories of software:
 - ▶ System software
 - ▶ Application software

System Software

Programs that control and manage the basic operations of a computer are referred to as **system software**, includes:

- ▶ **Operating System** controls the internal operations of the computer's hardware and manages all of the devices connected to the computer.
- ▶ **Utility Programs** perform a specialised task that enhances the computer's operation or safeguards data.
- ▶ **Software Developments Tools** are programs that are used to create, modify, and test software.

Application Software

Programs that people normally spend most of their time running on their computers, performing everyday tasks are referred to as **application software**.

- ▶ Word processing
- ▶ Spreadsheet
- ▶ Database
- ▶ Presentation

How Computers Store Data

How Computers Store Data

Concept

All data that is stored in a computer as sequences of 0s and 1s.

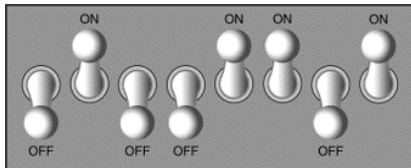


Figure: A byte as a sequence of eight switches

- ▶ A computer's memory is divided into tiny storage locations known as bytes
- ▶ One byte represents one number
- ▶ A byte is divided into eight smaller storage locations known as bits (binary digits)
- ▶ Bits are tiny electrical components that can hold either a positive or a negative charge.
- ▶ A positive charge is similar to a switch in the on position
- ▶ A negative charge is similar to a switch in the off position

Storing Numbers

- ▶ The positive charge or the on position is represented by the digit 1
- ▶ The negative charge or the off position is represented by the digit 0
- ▶ This corresponds to the binary numbering system where all numeric values are written as a sequence of 0s and 1s
- ▶ Each digit in a binary number has a value assigned to it

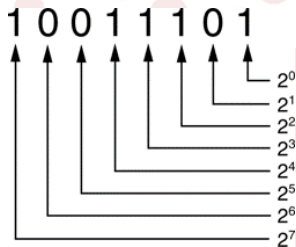


Figure: The values of binary digits as powers of 2

Binary number of 157

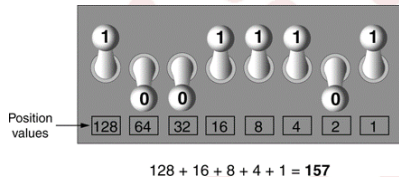
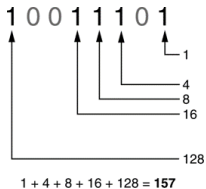


Figure: Determining the value of 10011101

Figure: The bit pattern for 157

- ▶ The largest value that can be stored in a byte is 255.
- ▶ Two bytes are used for larger numbers; maximum value is 65535.

Storing Characters

- ▶ Characters are stored in the computer's memory as binary number.
- ▶ **ASCII (American Standard Code for Information Interchange)** is a coding scheme.
- ▶ ASCII is a set of 128 numeric codes.
- ▶ ASCII is limited.
- ▶ **Unicode** is an extensive encoding scheme.
 - ▶ It is compatible with ASCII.
 - ▶ It represents characters for many languages in the world.

Advanced Number Storage

- ▶ Binary numbering system can be used to represent only integer numbers.
- ▶ Negative numbers are encoded using **two's complement**.
- ▶ Real numbers are encoded using **floating-point notation**.

Other Types of Data

- ▶ Digital data is data that is stored in binary.
- ▶ A digital device is any device that works with binary data.
- ▶ Digital images are composed of tiny dots of color known as pixels (picture elements).
- ▶ Digital sound is broken into small pieces known as samples.

How a Program Works

How do we communicate with computers?

- ▶ Computers' CPU can only understand **machine language** (0s and 1s).
- ▶ However, writing an entire program in machine language is very difficult and error-prone.
- ▶ Therefore, **programming languages** were invented.
- ▶ **Programming languages** allow us to write instructions in a form that computers can translate and execute.

Central Processing Unit (CPU)

CPU is designed to perform the following operations:

- ▶ Read a piece of data from main memory
- ▶ Adding two numbers
- ▶ Subtracting one number from another number
- ▶ Multiplying two numbers
- ▶ Dividing one number by another number
- ▶ Moving a piece of data from one memory location to another
- ▶ Determining whether one value is equal to another value

Fetch-decode-execute cycle is the term used when the CPU executes the instructions in a program.

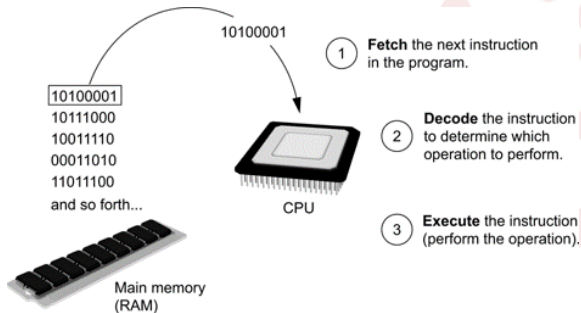


Figure: The fetch-decode-execute cycle

Machine Language and Assembly Language

- ▶ Computers only understand machine language.
- ▶ Machine language is difficult to write.
- ▶ **Assembly language** uses short words that are known as mnemonics.
- ▶ **Assembler** is used to translate an assembly language program to machine language.

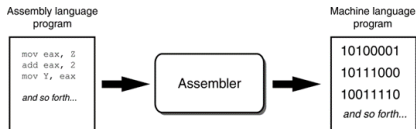


Figure: An assembler translates an assembly language program to a machine language program

High-Level Languages

- ▶ Assembly language is referred to as a low-level language.
- ▶ High-level languages allow you to create complex programs without knowing how the CPU works, using words that are easy to understand.

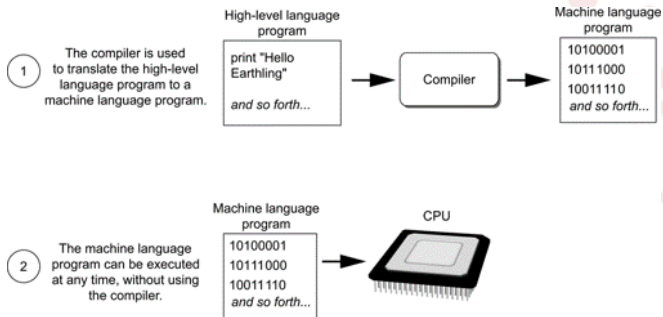
How a program works?

- ▶ The instructions written in a high-level programming language are called **source code** or simply **code**.
- ▶ Source code is translated to machine language using a **compiler** or an **interpreter**.

Compiler and Interpreter

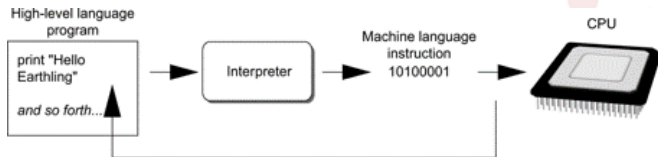
Compiler is a program that translates a high-level language program into a separate machine language program.

- Compiling a high-level program and executing it



Interpreter is a program that is a program that both translates and executes the instructions in a high-level language program.

- ▶ Executing a high-level program with an interpreter



The interpreter translates each high-level instruction to its equivalent machine language instructions and immediately executes them.

This process is repeated for each high-level instruction.

Getting Started: Python

Install Python

- ▶ Local installation: <https://www.python.org/downloads/>
- ▶ Jupyter Notebook: <https://jupyter.org/>
- ▶ Google Colab: <https://colab.google/>

The Python Interpreter

Python interpreter has two modes:

- ▶ Interactive mode: waits for a statement from the keyboard and executes it.
- ▶ Script mode: reads the contents of a file (Python script) and interprets each statement in the file.

Further Reading

- ▶ T. Gaddis, Starting out with Python. Harlow: Pearson Education Limited, 2022.

Any Question?